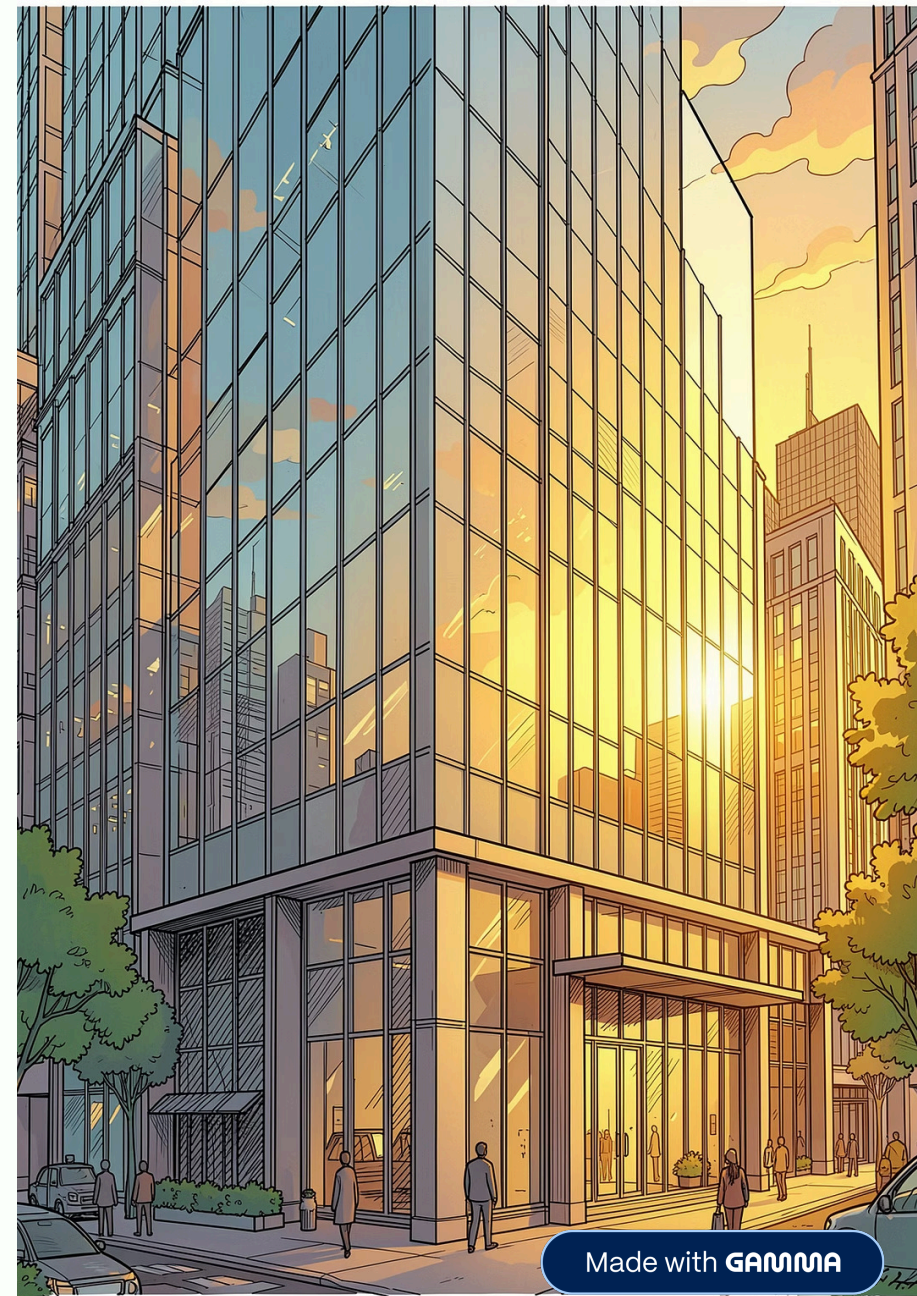


Bank Loan Analysis

A data-driven analysis of **38,576 consumer loan applications** using SQL and Power BI – uncovering loan performance, risk exposure, and borrower characteristics across a \$435.7M portfolio.



Dataset at a Glance

PostgreSQL was used to clean the data and calculate KPIs, then surfaced in an interactive three-page Power BI dashboard.

38,576

Loan Applications

Rows in the dataset

24

Columns

Borrower, loan & repayment fields

\$435.7M

Total Funded

Across all loan applications

\$473.0M

Amount Received

Total repayments collected

Overall KPIs

KPI	Total	MTD (Dec)	PMTD (Nov)	MoM Change
Loan Applications	38,576	4,314	4,035	▲ 6.9%
Total Funded Amount	\$435.7M	\$53.9M	\$47.7M	▲ 13.0%
Total Amount Received	\$473.0M	\$58.0M	\$50.1M	▲ 15.8%
Avg. Interest Rate	12.05%	12.36%	11.94%	▲ 3.5%
Avg. DTI	13.33%	13.67%	13.30%	▲ 2.8%

Good Loans vs. Bad Loans



Key Insight

Good Loans (Fully Paid + Current): **86.18%** of applications, \$370.2M funded, \$435.7M recovered.

Bad Loans (Charged Off): **13.82%** of applications, \$65.5M funded – only \$37.2M recovered.

⚠ The bank lost roughly **\$28M** on charged-off loans – funded \$65.5M but recovered only \$37.2M.

Loan Status Breakdown

Status	Applications	Funded	Received	Avg Rate
Fully Paid	32,145	\$351.4M	\$411.6M	11.64%
Charged Off	5,333	\$65.5M	\$37.3M	13.88%
Current	1,098	\$18.9M	\$24.2M	15.10%

Risk Pricing

Charged Off loans carry higher avg rates (13.88%) – riskier borrowers are priced higher.

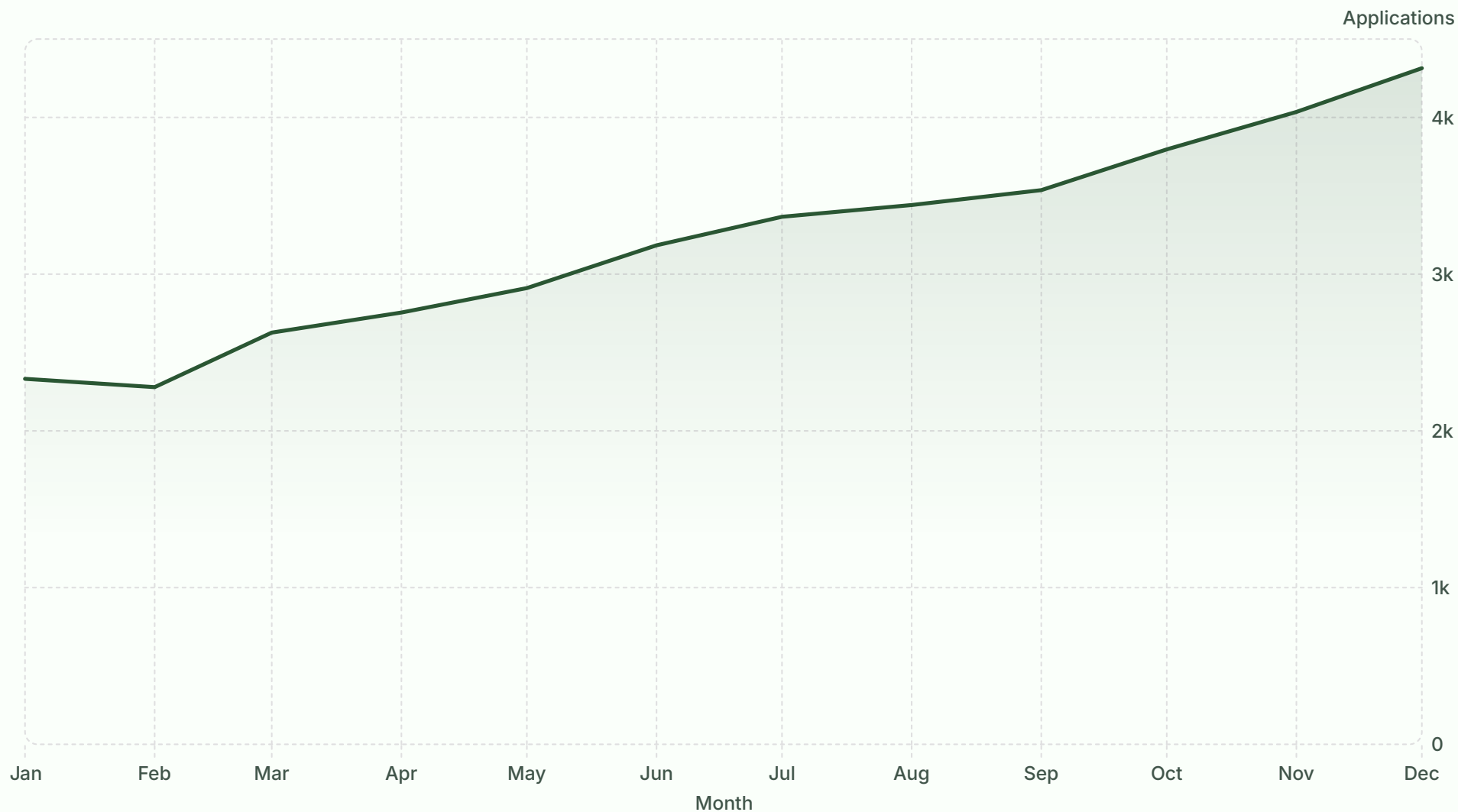
Active Loans

Current loans show the highest avg rate (15.10%), reflecting ongoing active payments.

Profitability

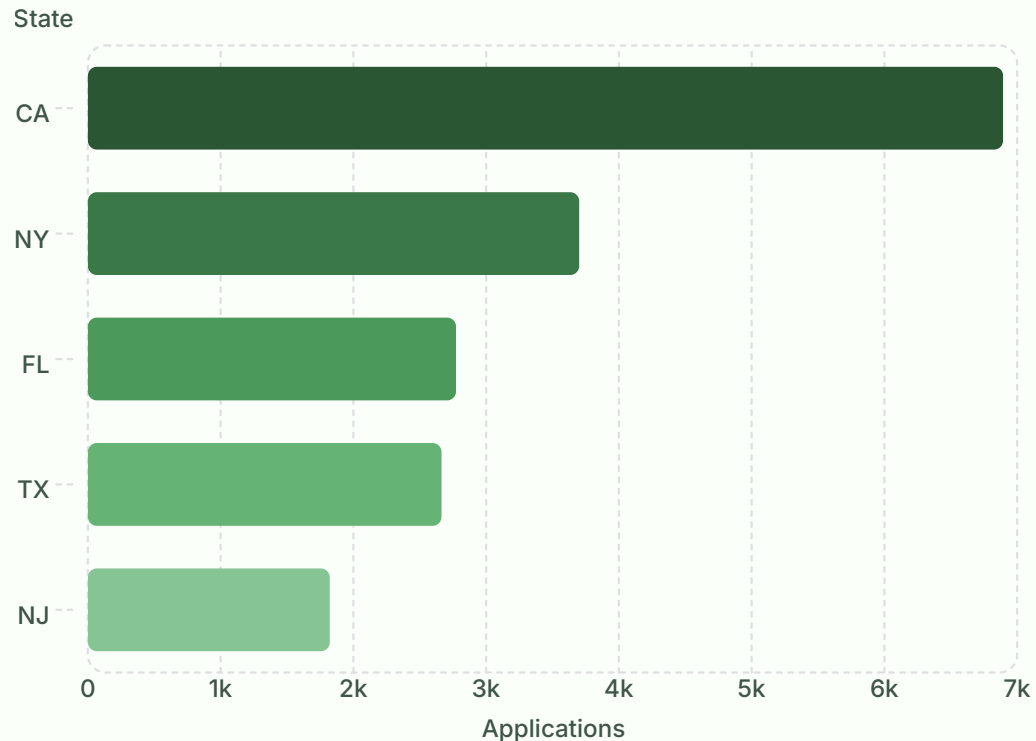
Fully Paid loans returned ~\$60M more than funded – healthy interest income.

Monthly Growth Trend



Applications grew steadily from **2,332 in January** to **4,314 in December** – nearly doubling over twelve months, with parallel growth in funded and received amounts.

Regional & Borrower Profile



Top States

California dominates with **6,894 applications** and \$78.5M funded – nearly double New York (\$42.1M). TX, FL, and NJ round out the top five.

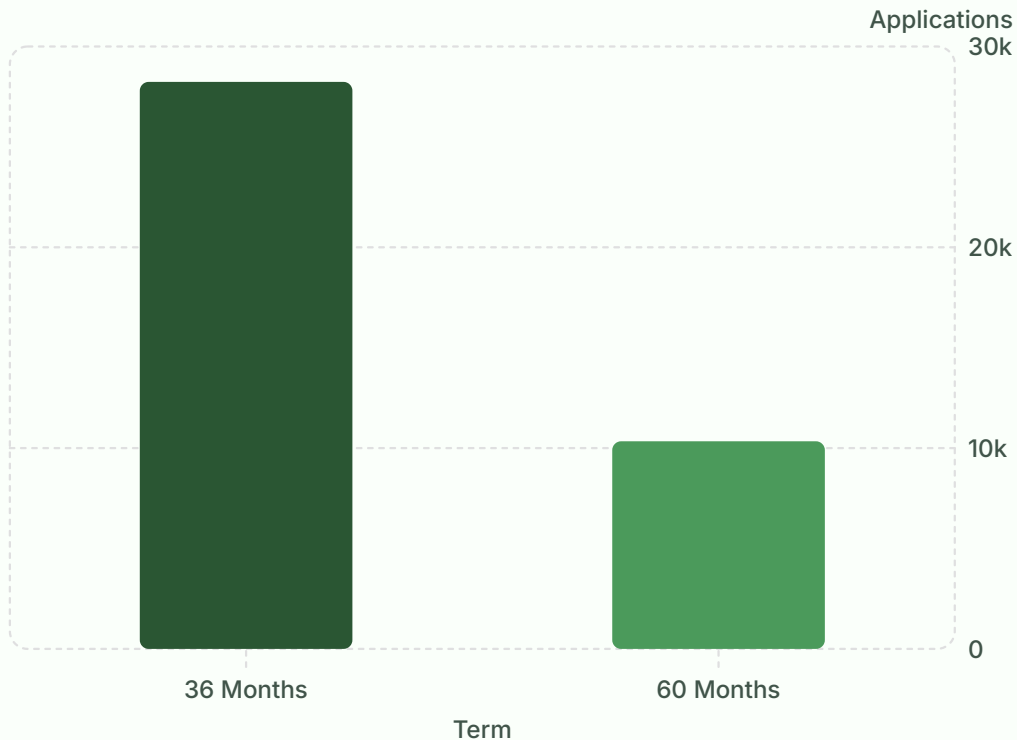
Employment Length

Borrowers with **10+ years** of employment account for the most loans (8,870) – a sign of more stable, established applicants.

Home Ownership

Renters (18,439) slightly outnumber mortgage holders (17,198) among applicants.

Loan Term & Purpose



Loan Term

36-month terms are far more common – **28,237 applications** vs. 10,339 for 60-month terms.

Top Loan Purposes

- **Debt consolidation:** 18,214 apps, \$232.5M funded
- **Credit card:** 4,998 apps, \$58.9M funded
- **Other:** 3,824 apps, \$31.2M funded
- **Home improvement:** 2,876 apps, \$33.4M funded

Debt consolidation is by far the leading purpose, followed by credit card refinancing.

Power BI Dashboard

A three-page interactive report with slicers for **Purpose**, **Grade**, and **State** driving every page.



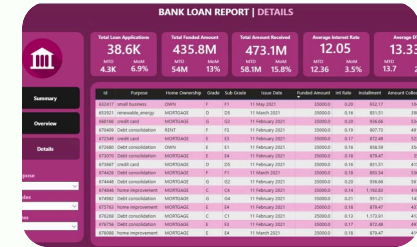
Summary Page

Headline KPIs, Good vs. Bad Loan breakdown, and Loan Status grid.



Overview Page

Visual breakdown by month, state, term, employment length, purpose, and home ownership.



Details Page

Row-level grid of individual loan records – purpose, grade, funded amount, rate, and collections.

Business Recommendations

→ **Reduce Bad-Loan Exposure**

13.82% charged off with ~\$28M shortfall – tighten DTI thresholds and grade-based limits for riskiest segments.

→ **Reward Stable Borrowers**

10+ years employment borrowers perform reliably – consider preferential rates or faster approval.

→ **Double Down on Top Purposes**

Debt consolidation and credit card refinancing dominate demand – tailor marketing and product bundling.

→ **Focus Regional Efforts**

CA, NY, TX, and FL drive bulk volume – prioritize regional marketing and local partnerships.

→ **Plan for Continued Growth**

Loan volume nearly doubled year-over-year – ensure underwriting and servicing capacity scales ahead.

→ **Watch Active Loans**

Current loans carry the highest avg rate (15.10%) – monitor this cohort closely as it migrates to Fully Paid or Charged Off.